

G11-FRONTIER-AUDIT-v1: Mapping remaining attack space - NO new branch yet

ZFL-CX

September 3, 2026

$G7 = FROZEN$, $G8 = CLOSED/BLOCKED$, $G9 BLOCKED$ (2 gates), $G10 CLOSED/BLOCKED$ (Ga

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}$

RULE : NO MÁS EXPLORACIÓN SIN NOVEDAD ESTRUCTURAL

1 Classes already killed

W_+ linear bounds, critical norms $L^3 \rightarrow X$, pressure, stretching algebra scale N_c^{-1} , concentration upper bound attempt, $S - \Omega$ scale, blow-up profile $r_n u$ compactness $r_n^{-1} E_0$, suitable-weak existence $\neq$ compactness, weak $\nrightarrow$ preservation of ε_0 , backward/unique continuation needing vanishing/smallness vs our lower $\geq \varepsilon_0$.

Diminishing returns inside E-G10 = brutal

2 Frontier audit - 3 candidate families

Candidate must pass:

new class $\wedge$ causal mechanism $\wedge$ connection to T^* $\wedge$ no reduction to E-G10

1. **Local energy / monotonicity** NOT CKN re-proved. Q: Is there monotone quantity $M(t)$ s.t. C_F^{conc} forces $M(t) \rightarrow +\infty$ incompatible with C_E^{weak} ? Need *concentration* $\rightarrow$ *monotonic* $\rightarrow \perp$ without assuming W_+ upper bound. Must be identity derivable from NS.

2. **Vorticity + topology** NOT spectral alignment. Q: Does global topology of vortex lines impose restriction incompatible with concentration $\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0 + N_c \rightarrow \infty$? Genuinely different from controlling W_+ . Would use C_J^{neg} loss of direction regularity as obstruction to singular knotting? Needs theorem linking topology to L^3 concentration.

3. **Rigidity of ancient via other route** G9 failed because we couldn't produce limit with uniform $L^2(B_R)$. But alternative procedure not using our direct critical compactness could produce ancient object classifiable. E.g., profile decomposition with extra tool, or contradiction via Liouville without $r_n^{-1} E_0$? Must not smuggle Type I.

Other: scale-critical quantity with opposite sign? Entropy?

3 Brutal table

Class	New?	Uses W_+ ?	New hyp?	Quant bridge?	Status
Local energy/monot.	?	<i>NO</i>	must not	needs $C_F \rightarrow M \rightarrow \perp$	AUDIT
Vorticity topology	?	<i>NO</i>	must not	needs geom $\rightarrow L^3$ incompat	AUDIT
Alt. ancient rigidity	?	<i>NO</i>	must not	needs new compact proc.	AUDIT
Others	?	?	?	?	AUDIT

Each row audited against: - Is it reducible to W_+ bound? If yes, discard (G7). - Is it reducible to new critical norm assumption? If yes, discard (G8). - Is it reducible to blow-up limit requiring $r_n^{-1}E_0$ uniform? If yes, discard (G9). - Is it reducible to vanishing/smallness not in $\mathcal{C}_{PROVEN}$? If yes, discard (G10).

Only row with explicit theorem + transfer from $\mathcal{C}_{PROVEN} + T^* < \infty \Rightarrow \perp$ can become G11-01.

Current: mapping, not choosing. No G11-01 yet.
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<i>G11-FRONTIER-AUDIT-v1 = DONE</i>

<i>STOP = PROGRESO</i> means we know where mechanism may hide, not that NS is solved
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